

Program Question_1

```
Implicit none
Real::x,res
!Interpolated value
x=8.0
Call ForwardDifference(x,res)
Write(*,*)"Estiamate the river's water level at 8:00 am",x,"is",res
contains
Subroutine ForwardDifference(x,res)
  Implicit none
  Real,Intent(in)::x
  Real,Intent(out)::res
  Real,Dimension(6)::time,level
  Real,Dimension(6,6)::delta
  Real::u,h
  Integer::n,i,j
  !Initialize value
  time=[2.0,6.0,10.0,14.0,18.0,22.0]
  level=[1.2,1.5,1.9,2.3,2.1,1.8]
  n=6
  delta(:,1)=level
  Do j=2,n
    Do i=1,n-j+1
      delta(i,j)=delta(i+1,j-1)-delta(i,j-1)

    End Do
  End Do
  !stepsize
  h=time(2)-time(1)
  u=(x-time(1))/h
  res=level(1)
  Do j=2,n
    res=res+(u*delta(1,j))/factorial(j-1)
    u=u*(u-(j-1))
  End Do
End subroutine
Integer Function factorial(n)
  Integer,Intent(in)::n
  Integer::i
  factorial=1
  Do i=1,n
    factorial=factorial*i
  End Do
end Function
```

End Program